

Course Project Design and Datasets

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Outline

- Course project workflow and challenges
- Choose a topic and find datasets
- Agenda of project process
- Final deliverables and grading

Project workflow



1 bd | 1 ba | 766 sqft
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Example:
Predicting house
sale prices

Collect house sale
records in the past,
features of houses

Train a regression
model

Evaluate the
performance,
analyze results

Problem
Formulation

Collect and
Process Data

Train and
Tune Models

Evaluation
and Analysis

Challenges and time distribution

Formulate Problem

- Convert a problem into a ML task

Data

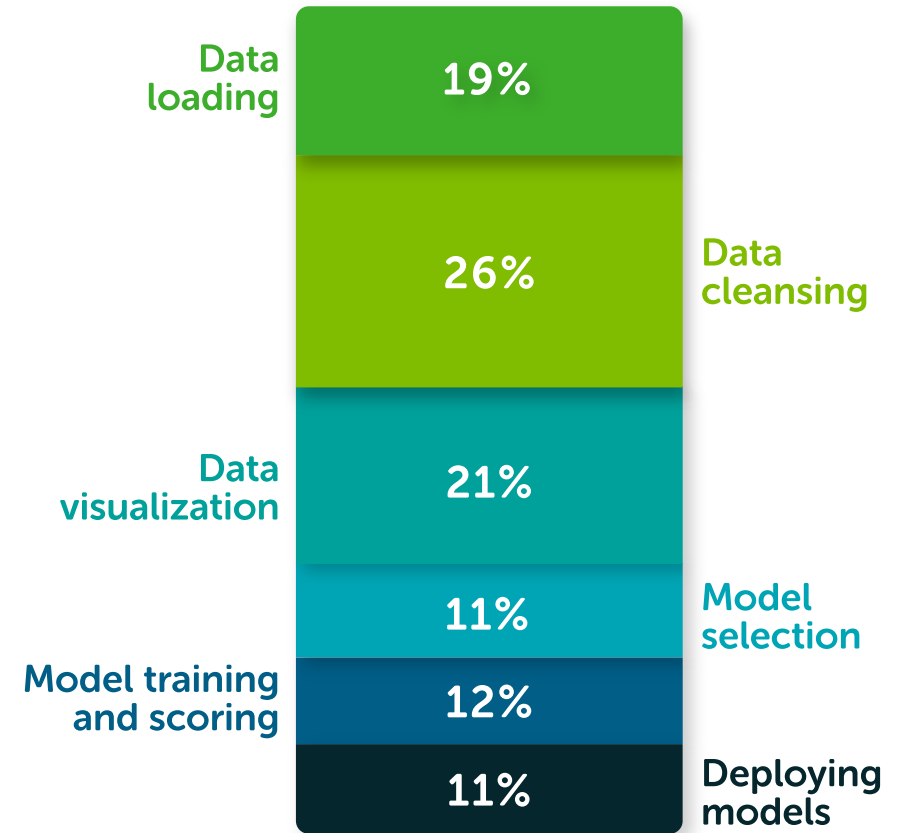
- Data collection, data exploration and preprocessing

Train Models

- Model selection, optimization, computation

Evaluation and Analysis

- Evaluation metrics, baseline, insight analysis



How data scientists spent their time?

45% of their time is spent getting data ready

21% of their time on data visualization

Outline

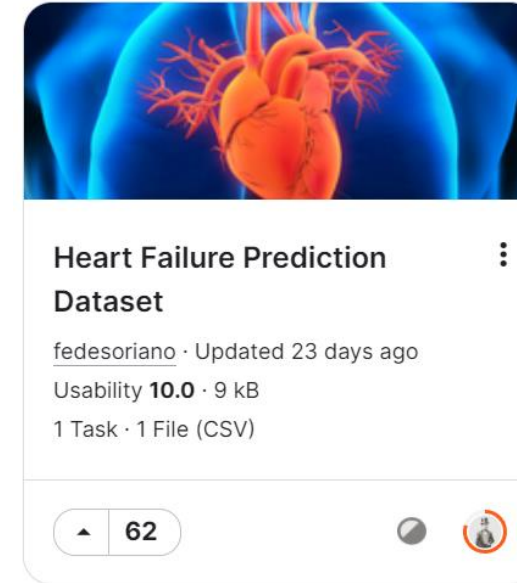
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Design a ML project

- Choose a topic you are interested to explore
 - Depend on the data you have, the area you want to work on
 - The problem is practical, analytical, and makes sense (use google)
 - The target (the things you want to predict) can be quantified and correlated with your features (use common sense)
 - The performance can be evaluated with certain metrics. The results can be reproduced
 - The workload is manageable – 2 months for this course project. Don't work on a problem which is too hard or a dataset which is too complex

Where to find datasets

- [Paperswithcode Datasets](#)
 - Academic datasets with leaderboard
- [Kaggle Datasets](#)
 - ML datasets uploaded by data scientists
- [UCI Machine Learning Repository](#)
 - Open-source academic datasets
- [Google Dataset search](#)
 - Search datasets in the Web
- Toolkits datasets
 - [tensorflow](#), [huggingface](#), [scikit-learn](#), [Yahoo finance](#)
- [Open Data on AWS](#)
 - 100+ large-scale raw data
- Government open data
 - [New York City](#), [Denver](#)



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Agenda (1)

- Form a team (**3-5 students**), utilize canvas to find teammates
- Proposal (**due TBD**)
 - Pick a project idea to work on
 - Title, project team members (name and ID), **300-500 words** description of what you plan to do
 - Motivation: What problem are you tackling?
 - Data: What data are you learning from?
 - Method: What techniques are you planning to apply? What challenge are you trying to solve?
 - Experiments: What experiments are you planning to run? How do you plan to evaluate your framework?

Agenda (2)

- Final Report (**due TBD**)
 - At most **10 pages**
 - **1. Abstract (optional) and Introduction**
 - Motivation for pursuing this problem
 - Background of the problem
 - What is the input and output of your framework (explicitly with features and predictions)
 - What is the models you use and why
 - What is the conclusion and insights you have got
 - **2. Related Work**
 - What is the challenge of your study problem?
 - What is the existing method for your study problem (papers, online search results, less than 5)
 - What is their strength and weakness
 - How is your work similar or different from them?

Agenda (3)

- Final Report (**due TBD**)
 - **3. Dataset and Formation**
 - Describe your dataset, how it is collected, feature list, data statistics (e.g., feature/target distribution, male/female, number of subjects)
 - What is the data preprocessing you did (normalization, missing value, feature selection, sampling ...)?
 - What are the inputs and output?
 - What is the data size (training/validation/testing)?
 - **4. Method**
 - Your learning algorithms
 - Describe the model, its mechanism, how it works in your scenario
 - Learning objective in formal language and math equations (refers to textbook and lecture notes)
 - What is the strength and weakness of your model?

Agenda (4)

- Final Report (**due TBD**)
 - **5. Experiments:**
 - What is the setting of your experiments and how you find it (settings for your model like learning rate, iteration, batch size, cross-validation)
 - Describe your training procedure
 - Analyze if you had overfitting problem and how to handle it
 - What are your baselines and why?
 - List and explain your evaluation metrics (accuracy, precision, AUC...)
 - Show results in tables and visualization
 - Show qualitative insights from your results
 - Discuss your results and the future work (optional)
 - **6. Conclusion:** what have you done, what did you find (short)
 - **7. References:** The existing work you had reviewed (less than 5)

Suggestions

- Quick proof-of-concept
 - Use small amounts of data and simple model to validate your idea
 - The simple model can be used as baseline in the future
- Developing and report simultaneously
 - Start your report when developing idea and doing experiments
 - Writing reports can help you get clear on what you need to do and organize your results
- Control the size of your project
 - It is a 2-month course project, don't deal with too large datasets, too many features and too complex problems
 - You always have the chance to keep working on your project after the course, and develop it into research topics
- Use LLMs for idea, method, and result analysis
 - ChatGPT can do lots of things more than you can imagine – e.g., learning guidance, data augmentation, visualization.

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Final deliverables and grading

- Report
 - In PDF, with clear title, author names, and KU ID
 - All the sections we discussed
 - Contribution of team members, description of working contents
- Supporting materials
 - Codes
 - Documentation include explanation of code sections, and screenshots of code outputs.
 - In zip or link to github, don't exceed 5MB
- Project Grading
 - Clarity of the report
 - The relevance of the project to topics taught in the class
 - The novelty of the problem
 - The technical quality and significance of the work

Where to Find Project Ideas

- <https://cs229.stanford.edu/proj2020spr/>
- <https://medium.com/coders-camp/180-data-science-and-machine-learning-projects-with-python-6191bc7b9db9>
- <https://data-flair.training/blogs/machine-learning-project-ideas/>